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HEADQUARTERS
COUNTER INTELLIGENCE CORPS REGION I
970TH COUNTER INTELLIGENCE CORPS DETACHMENT
EUROPEAN COMMAND

FWM/ola

APD 154

File No: I-1606

16 January 1948

MEMORANDUM FOR THE OFFICER IN CHARGE

SUBJECT: HORTEN, Reiner and Walter (brothers)

Re: Operation HARASS

1. Reasons for Investigation

a. Reference is made to letter, Headquarters, 970th CIC Detachment, File: D-198239, Subject: Flying Saucers, dated 28 October 1947, paragraphs 2-a and 2-b.

b. Reference MOIC, Headquarters, Sub-Region WUERTTEMBERG, CIC Region I, Subject as above, File: I-W-D-775, dated 24 November 1947.

c. Reference Air Intelligence Guide for Alleged "Flying Saucer" Type Aircraft (written at Wright Field, Ohio).

d. Reference the magazine DER FLIEGER, April 1943, Publisher: Walter ZUERL, MUNICH.

e. The following information was obtained from an informant as yet uncarded.

f. Paragraph 2, following, is lettered in an unorthodox manner in that sub-paragraph letters are repeated. This lettering system is used in order to correspond exactly with the lettering used for the questions in the KEI, reference 1-c above. All information obtained is arranged to specifically answer as many of the KEI's as possible. In reading this report, it would be of assistance to have the KEI's at hand.

2. Results of Investigation

a. Scientists who have a better than average knowledge of the HORTEN brothers' work are:

(1) PRANDTL, Prof. Ludwig, [REDACTED] (L52/U52).

(2) LIPPISCH, Prof., fm, [REDACTED] U.S.A.
FREDMANN

(3) FREIMANN, fm, Diplom Ingenieur, worked as assistant to the HORTEN brothers. He is last known to have lived in BAD CANSTATT (L49/S12). He is reported to have contacted TECHNICAL INFORMATION CENTER, G-2, Personnel Branch, in MUNICH (M49/Y85), in February 1947. Since that time no information concerning him has been obtained. Description of FREIMANN is as follows:

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Height: 5'-9" to 5'-11"
Weight: 150 to 155 lbs.
Hair : Dark brown
Eyes : Brown
Face : Long, oval

b. - e. Unknown.

f. Present whereabouts of the HORTEN family is unknown, but possibly in
BONN (K51/P53) which was the HORTEN's home.

g. Unknown.

h. (1) According to informant, the first four (4) HORTEN models were
gliders, the fourth having been a powered glider. From these glider developments the
parabolic wing plan was evolved. Description and illustrations of the HORTEN I, II,
III and IV may be seen in the April 1943 issue of the magazine DER FLIEGER, Pub-
lisher: Walter ZUERL, MUNICH.

(2) The HORTEN models which followed culminated in a rocket powered,
supersonic interceptor. Only the prototype of this last ship was built. Plan of
this ship is shown in Exhibit A. This plane carried only one man who flew the ship
lying on his stomach. Cruising speed was 900 kilometers per hour, with a flight
duration of thirty five (35) minutes. Fullpower could be turned on for a one-minute
burst and a speed of 1600 kilometers per hour was obtained. At this point the fuel
was exhausted and the plane glided down to a landing. Maximum altitude obtained was
sixteen (16) to seventeen (17) kilometers. These performances were achieved in 1944.
(Agent's Note: Informant did not witness these record flights and cannot give the
name of anyone who did. He claims, however, that these performances were "generally
known" among aeronautical engineers. These flights may possibly have occurred near
GOETTINGEN as most of the development of this plane was at GOETTINGEN.) Informant
does not know where this plane is today or what happened to it.

(3) This ship had no vertical rudder. Under gliding conditions
three (3) sets of wing-controls controlled the ship. One (aileron) set banked and
turned the plane. The second set operated as elevators for climbing. The third set
operated as elevators for nosing over and diving. A flap on the leading edge also
assisted the control. Under powered flight the gliding controls were locked, and
control was obtained by varying the directions of the two rockets (rotation in a
vertical plane only).

Questions Concerning an Oval Shaped Aircraft

a. Wing cross section was as noted in Exhibit C with a maximum thick-
ness at about 2/3 of the chord from the leading edge. Boundary layer control was
achieved by sucking air into the wing near the trailing edge as shown in the illus-
tration. Vents were spring loaded to allow only a suction. Suction was produced
by a pump for this purpose. This control was successful in smoothing out heavy
turbulence.

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- b. Controls - See h-(3) above.
- c. There were no ^{openings} ~~spaces~~ in the leading edge for boundary layer control.
- d. Airfoil shape was as shown (Exhibits A and C). This airfoil was designed for supersonic performances. Ratio of maximum thickness to chord was approximately 1/10 at the root of the wing. Near the tips this ratio was somewhat less but the section was similar.
- e. A front view of the plane is shown in Exhibit B.

Items of Construction

- a. - c. Construction was of conventional aluminum alloy type. Wing beams were of steel. Parabolic wing plan with rockets at the wing tips was considered to be a very strong structure since stresses tended to straighten out the parabola.

Items of Arrangement

- a. Pilot visibility was considered good since the pilot's head was well forward and considerable plexiglass was used.
- b. Crew number: one (1) - lying on his stomach.
- c. Pressurized cabin equipment: No details given.
- d. Escapement method: The canopy was easily disconnected and the pilot was thrown clear of the plane by a spring system.
- e. All accessories were driven by auxiliary power plants (one in front of each rocket). The auxiliary plants were turbines driven by the decomposition of hydrogen peroxide. Exhaust from these turbines furnished a fraction of the oxygen for the rockets.
- f. There were no provisions for towing or for refueling in flight.
- g. Assistance was not required for take-off.
- h. Bombay provisions: No details given, possibly had none. This plane was an interceptor.

Landing Gear

- a. - d. Take-off was by means of a six-wheeled carriage on which the plane was set. When the ship took off, the carriage was left on

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the ground. Take-off run was about one (1) kilometer. Landing shock was taken up by skids built into the fuselage. Landing speed was about 250 kilometers per hour. Ice, snow or water take-offs were not attempted.

Power Plant

a. - e. Power was obtained from rockets at the two wing tips. Fuel burned was methyl alcohol. Liquid oxygen at sixty (60) atmospheres pressure supported the combustion. This oxygen supply was increased by the exhaust from the auxiliary turbines. The oxygen was preheated in the jacket of the rocket before entering the combustion chamber. Under normal cruising conditions the two rockets gave a total thrust of fifteen hundred (1500) pounds. Maximum total thrust for the two rockets was eight thousand (8000) pounds. It is believed that no special catalytic agent was used to assist combustion.

3. Agent's Comments and Recommendations

a. Information above is evaluated F-3.

b. Informant seems to understand what he is talking about; however, he is still quite young and does not have an engineering degree. He is self-educated in part, and has reached an advanced stage in some fields, at least. He has done engineering work in connection with rocket research and claims to know the HORTEN brothers. He also claims to have seen the HORTEN brothers' plane in flight. He made no statements which the agent could say were technically false.

c. Suggest that PRANDL and LIPPISCH, mentioned in paragraph 2-a above, be contacted on this case. Neither of the men is in the U.S. Zone of Germany. Suggest that Region I attempt to locate FREIMANN, also mentioned in paragraph 2-a. Suggest that a copy of this report be forwarded to Region IV for a lead on FREIMANN. It is possible that FREIMANN may have reported to TIB in MUNICH.

d. Lead in paragraph 2-f suggests that the HORTEN's, or their relatives may be in ROSEN (British Zone).

e. Lead in paragraph 2-h-(2) suggests that further leads might be developed in GOETTINGEN (British Zone).

APPROVED:

HERMAN H. KROH
Special Agent CIC
Operations Officer

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Fred W. Matting
FRED W. MATTING
Special Agent, CIC

Incls: 1. Exhibit A - Plan view of HORTEN airplane
2. Exhibit B - Front view of HORTEN airplane
3. Exhibit C - Detail view of aircraft section

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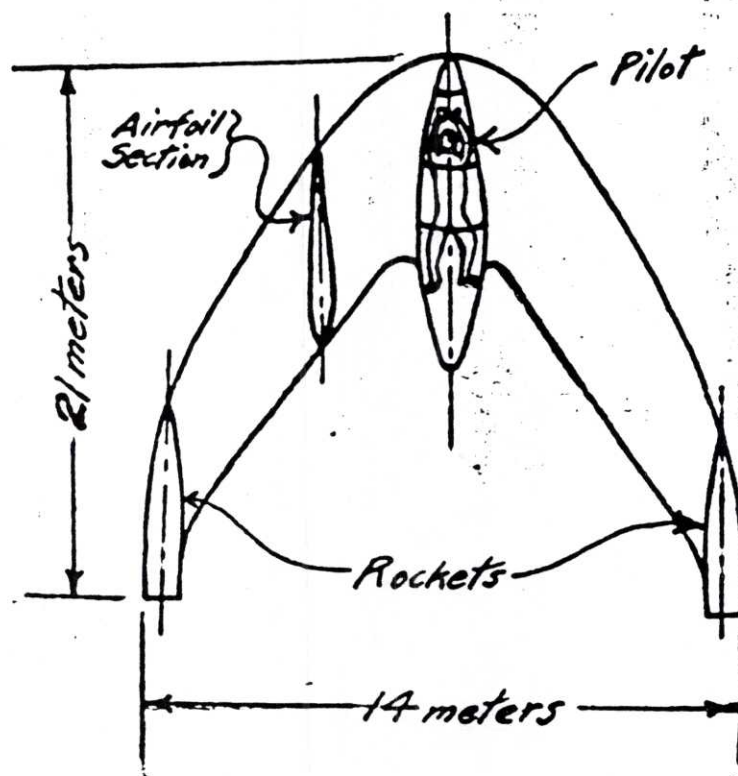
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Plan View of HORTEN
Airplane

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Exhibit A

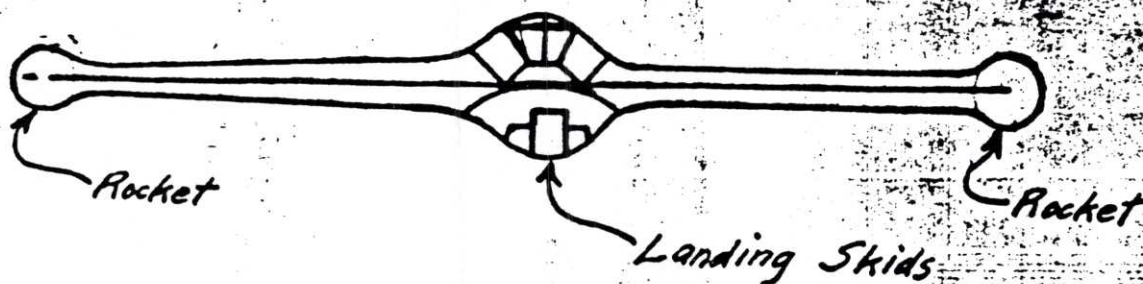
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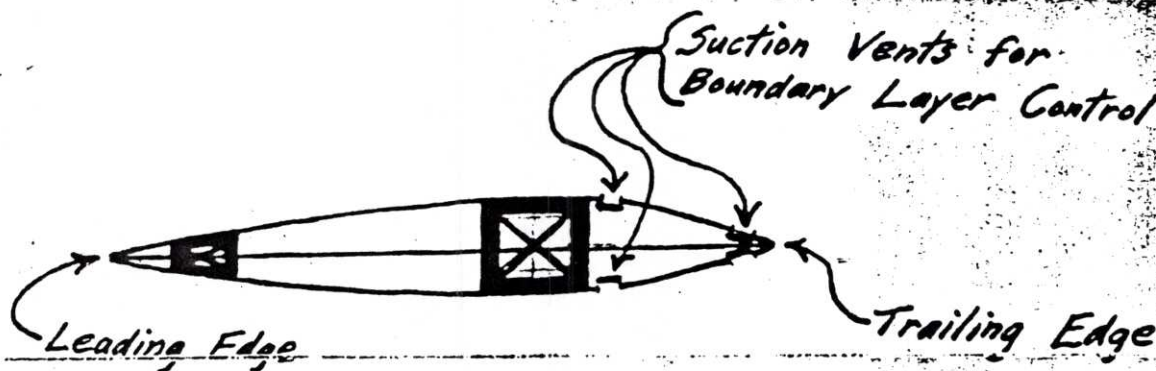
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Front View of HORTEN
Airplane

Exhibit B



Airfoil Section
(Becomes Relatively Thinner Near Wing Tips)

Exhibit C

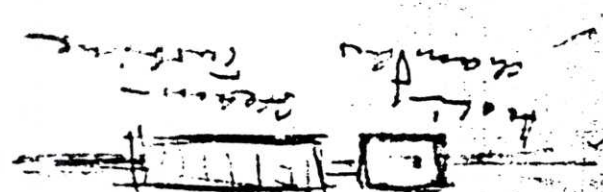
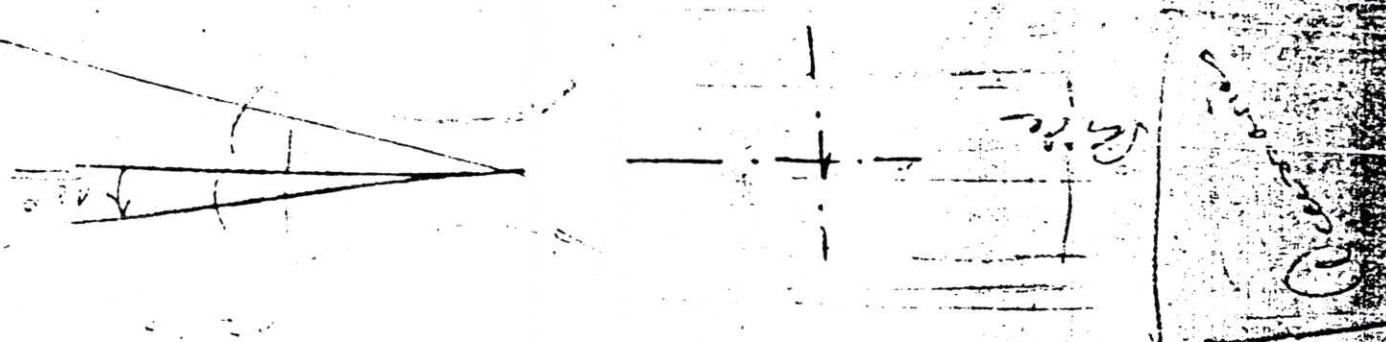
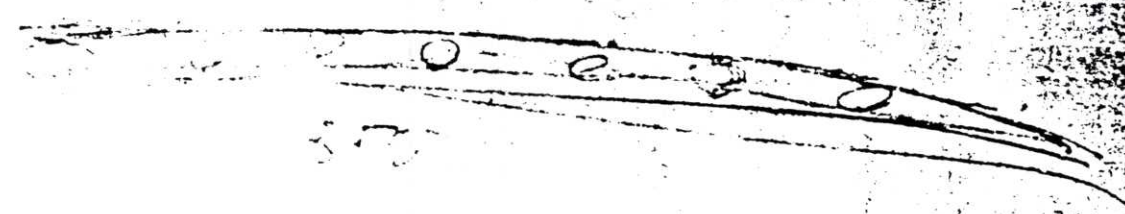
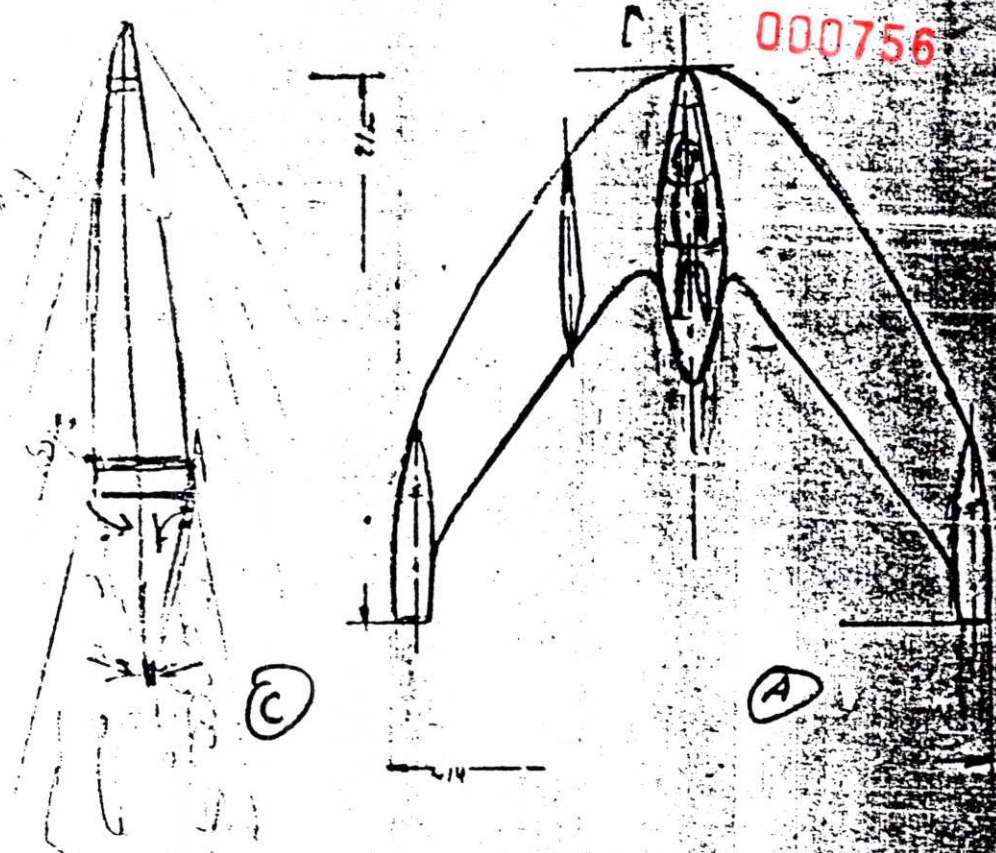
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